Frame conditions from schematic axioms (partially functional, functional, weakly dense)

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the iftag{prvBox}{}{...} branches carry the lem:box-iff-diamond citations in the else group - rendered in full once the preamble iftag branch semantics is fixed (tag-macro fix takes the active group, else group needed); item 2's pointers are to item 1 (same proof) and thm:completeframeprops.

Editorial note. The label prop:anotherfive-a names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Box-diamond interplay in maximal consistent sets. Same unit, \ollabel{lem:box-iff-diamond} (registered card). Converts between Box-membership and Diamond-membership when reading R^Sigma from set inclusions.
- Lindenbaum extension. Same unit, \ollabel{thm:lindenbaum} (registered card). Item 3: extends the constructed Sigma-consistent set Gamma to a complete maximal consistent set Delta_3.

Source reference. Original theorem thm-completeframeprops: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 675–675. Prerequisite pointer invoked at proof line(s) 822; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference tab-anotherfive: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 798–798. Named citation invoked at source line(s) 798: the author's marker \olref[frd][acc]{tab:anotherfive} resolves inside the full Open Logic Project to another unit (not staged in this

corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma lem-box-iff-diamond: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1247–1247. Prerequisite pointer invoked at proof line(s) 812; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma lem-box1: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1155–1155. Prerequisite pointer invoked at proof line(s) 868; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-derivabilityfacts: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 855–855. Named citation invoked at source line(s) 855: the author's marker \olref[prf][prp]{prop:derivabilityfacts} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-derivabilityfacts-deduction: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 855–855. Named citation invoked at source line(s) 855: the author's marker \olref[prf][prp]{prop:derivabilityfacts-deduction} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-derivabilityfacts-monotonicity: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 875–875. Named citation invoked at source line(s) 875: the author's marker \olref[prf][prp]{prop:derivabilityfacts-monotonicity} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let Σ be a normal modal logic; then:*

1. *If Σ contains the schema $\Diamond A \to \Box A$ then the canonical model for Σ is partially functional.*
2. *If Σ contains the schema $\Diamond A \leftrightarrow \Box A$ then the canonical model for Σ is functional.*
3. *If Σ contains the schema $\Box\Box A \to \Box A$ then the canonical model for Σ is weakly dense.*

(see tab-anotherfive for definitions of these frame properties).

Proof. 1. Suppose that Σ contains the schema $\Diamond A \to \Box A$, to show that R^Σ is partially functional we need to prove that for any Δ_1, Δ_2, $\Delta_3 \in W^\Sigma$, if $R^\Sigma\Delta_1\Delta_2$ and $R^\Sigma\Delta_1\Delta_3$ then $\Delta_2 = \Delta_3$. Since $R^\Sigma\Delta_1\Delta_2$ we have $\Box^{-1}\Delta_1 \subseteq \Delta_2$ and since $R^\Sigma\Delta_1\Delta_3$ also $\Box^{-1}\Delta_1 \subseteq \Delta_3$. The identity $\Delta_2 = \Delta_3$ will follow if we can establish the two inclusions $\Delta_2 \subseteq \Delta_3$ and $\Delta_3 \subseteq \Delta_2$. For

the first inclusion, let $A \in \Delta_2$; then $\Diamond A \in \Delta_1$, and by the schema and deductive closure of Δ_1 also $\Box A \in \Delta_1$, whence by the hypothesis that $R^\Sigma \Delta_1 \Delta_3$, $A \in \Delta_3$. The second inclusion is similar.

2. This follows immediately from part 1 and the seriality proof in thm-completeframeprops.

3. Suppose Σ contains the schema $\Box\Box A \to \Box A$ and to show that R^Σ is weakly dense, let $R^\Sigma \Delta_1 \Delta_2$. We need to show that there is a complete Σ-consistent set Δ_3 such that $R^\Sigma \Delta_1 \Delta_3$ and $R^\Sigma \Delta_3 \Delta_2$. Let:

$$\Gamma = \Box^{-1}\Delta_1 \cup \Diamond\Delta_2.$$

It suffices to show that Γ is Σ-consistent, for then by Lindenbaum's Lemma it can be extended to a complete Σ-consistent set Δ_3 such that $\Box^{-1}\Delta_1 \subseteq \Delta_3$ and $\Diamond\Delta_2 \subseteq \Delta_3$, i.e., $R^\Sigma \Delta_1 \Delta_3$and $R^\Sigma \Delta_3 \Delta_2$ (by lem-box-iff-diamond).

Suppose for contradiction that Γ is not consistent. Then there are formulas $\Box A_1$, ..., $\Box A_n \in \Delta_1$ and B_1, ..., $B_m \in \Delta_2$ such that

$$A_1, \ldots, A_n, \Diamond B_1, \ldots, \Diamond B_m \vdash [\Sigma]\bot.$$

Since $\Diamond(B_1 \wedge \cdots \wedge B_m) \to (\Diamond B_1 \wedge \cdots \wedge \Diamond B_m)$ is derivable in every normal modal logic, we argue as follows, contradicting the consistency of Δ_2:

$A_1, \ldots, A_n, \Diamond B_1, \ldots, \Diamond B_m \vdash [\Sigma]\bot$

$A_1, \ldots, A_n \vdash [\Sigma](\Diamond B_1 \wedge \cdots \wedge \Diamond B_m) \to \bot$

by the deduction theorem

prop-derivabilityfactsprop-derivabilityfacts-deduction, and TAUT

$A_1, \ldots, A_n \vdash [\Sigma]\Diamond(B_1 \wedge \cdots \wedge B_m) \to \bot$

since Σ is normal

$A_1, \ldots, A_n \vdash [\Sigma]\neg\Diamond(B_1 \wedge \cdots \wedge B_m)$

by PL

$A_1, \ldots, A_n \vdash [\Sigma]\Box\neg(B_1 \wedge \cdots \wedge B_m)$

$\Box\neg$ for $\neg\Diamond$

$\Box A_1, \ldots, \Box A_n \vdash [\Sigma]\Box\Box\neg(B_1 \wedge \cdots \wedge B_m)$

by lem-box1

$\Box A_1, \ldots, \Box A_n \vdash [\Sigma]\Box\neg(B_1 \wedge \cdots \wedge B_m)$

by schema $\Box\Box A \to \Box A$

$\Delta_1 \vdash [\Sigma]\Box\neg(B_1 \wedge \cdots \wedge B_m)$

by monotonicity, prop-derivabilityfactsprop-derivabilityfacts-monotonicity

$\Box\neg(B_1 \wedge \cdots \wedge B_m) \in \Delta_1$

by deductive closure;

$\neg(B_1 \wedge \cdots \wedge B_m) \in \Delta_2$

since $R^\Sigma \Delta_1 \Delta_2$.

□